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Feasibility study on synthermal storage of two cryogenics in a single tank with a metal common bulkhead

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Abstract. Liquid-oxygen-liquid-methane reusable rocket technology is currently a hot topic in the field of spaceflight. Storage of liquid methane and liquid oxygen in a single tank but with a common bulkhead partition is the most compact and favorable scheme. Usually, two cryogenics with different saturation temperatures are separated by a partition made of some low-conductivity materials or insulation structures, which creates difficulties in the processing and manufacturing of the tank. Instead of keeping the pressure on bilateral compartments close to each other, an idea is proposed to have the partition made of the same metal as the tank wall, which turns out a thermal equilibrium result inside both compartments. To explore the feasibility of the new method, a small-scale storage system with a stainless-steel common bulkhead tank is designed and tested by measuring the transient thermodynamic behavior inside the tank during the self-pressurization and venting processes. Experiments show that the pressures in the liquid oxygen and liquid methane compartments are 399.36 kPa and 67.86 kPa respectively (Scheme 1), which means a large pressure difference between the two compartments, and even worse a subatmospheric state in the liquid methane compartment. Consequently, appropriate amount helium is charged into the liquid methane compartment to improve the performance (Scheme 2). The results indicate that Scheme 2 significantly prolongs the storage duration, with a 38.30% increase in self-pressurization and venting time of the liquid oxygen compartments, and a 9.70% increase on the liquid methane compartment, compared with Scheme 1 with the same filling ratio and boundary conditions.

1. Introduction

Cryogenic propellants are the preferred choice for reusable launch vehicles due to their high specific impulse, precise controllability, and environmental friendliness [1]. In the past decades, cryogenic bipropellants have been extensively studied and widely used in space missions [2, 3].

Common bulkhead (CBH) tank acts as a promising solution for lightweight and efficient bipropellant storage. In a CBH tank, the fuel and the oxidizer are separated by a common bulkhead partition. Singh et al. [4] evaluated the performances of different launch vehicles utilizing either sandwich CBH tanks or individual tanks. They found that a possible reduction in total weight and height of these vehicles by at least 490 kg and 1.755 m, respectively. To develop a long-term common bulkhead storage, it is crucial to reveal the relationship between the thermal behaviors and the storage duration [5, 6]. Self-pressurization behavior is an unavoidable issue in a cryogenic



storage tank. Hence, cryogenic fluids stored in tanks must be vented when the storage pressure exceeds the maximum allowable pressure of the tank. Extensive research has been conducted on investigating self-pressurization and venting behaviors in individual tanks [7-9]. However, few studies have focused on CBH tanks.

Therefore, a nonadiabatic CBH storage concept is employed to store liquid methane (LCH_4) and liquid oxygen (LOX). Considering the thermophysical properties of oxygen and methane, common bulkhead storage experiments are conducted with the presence and in absence of helium gas inside the LCH_4 compartment. The storage states and transient thermal behaviors during self-pressurization and venting processes are analyzed.

2. Experiment apparatus and methodology

An experimental setup for common bulkhead cryogenic fluids storage was established, which allows evaluating the effect of helium gas on the thermal behaviors and storage durations of LOX and LCH_4 in a CBH tank. The main components of the experimental setup include a vacuum chamber, a liquefaction subsystem with a liquid nitrogen (LN_2) trough, a tank with a stainless-steel common bulkhead, gas charging and venting units, valves, sensors, and data acquisition devices. The pressurant helium gas is introduced into the LCH_4 compartment through the filling pipeline of LCH_4 at a pressure of 2 bar and at room temperature. Figure 1 presents the photo of all the system components.

The control and data acquisition unit includes eighteen four-wire platinum resistance thermometers, two high-precision pressure transducers, three mass flowmeters, two solenoid valves, a temperature controller, an automatic thermostat device, and a multichannel data acquisition unit. Five thermometers are placed along a vertical thin stainless-steel rod at a 15 mm interval in the LCH_4 compartment. Seven thermometers are placed along a vertical thin stainless-steel rod at a 15 mm interval in the LOX compartment. The diameters of both rods are 1 mm. Six thermometers are mounted on the upper and lower surfaces of the CBH. These thermometers are used to monitor the temperature distributions and the liquid levels. The distribution of thermometers is depicted in Figure 2. The pressures in the CBH tank compartments are monitored by Gefran pressure transducers. The pressure data is compared with the pre-set range of pressure control band to control the on/off state of solenoid valves. The solenoid valve is turned on when the pressure in the compartment exceeds the higher limit of the pressure control band and is turned off when the pressure is below the lower limit. The mass flow rates of the exhausted oxygen and methane are measured using two Alicat® mass flowmeters. A Lake Shore® 335 temperature controller along with a four-wire PT-100 resistor and an electric heater is introduced to compensate the heat leakage from the LN_2 trough. All signals in the system are acquired by a

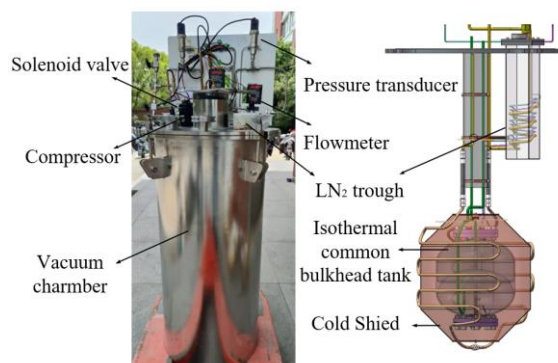


Figure 1. Photo of experimental setup facility.

Keysight DAQ970A device and recorded by a LabVIEW program. The measured parameters and the uncertainties of the sensors and equipment used in the system are listed in Table 1. To investigate the self-pressurization and venting behaviors of oxygen and methane, two schemes are conducted. Their operation conditions are shown in Table 2.



Figure 2. Distribution of thermometers in the CBH tank. (a) Thermometers in the LOX and LCH₄ compartments. (b) Thermometers on the CBH surface.

Table 1. List of measured parameters and main measurement instruments.

Parameters	Type	Range	Uncertainty
Temperature	Platinum resistor	77 K-400 K	± 0.1 K
Pressure	Gefran	0 kPa-600 kPa	0.25% FS
Mass flow rate	Alicat®	0 SLPM-250 SLPM	$\pm 0.8\%$ of reading and $\pm 0.2\%$ FS
CBH temperature	LakeShore model335	4 K-400 K	± 0.05 K

Table 2. Operation conditions of storage experiments.

Scheme	The filling ratio	Thermal boundary condition	Pressure control band (PCB)
No-helium	LOX: 90%	303 K	PCB for LOX: 500 ± 50 kPa
Helium-mixing	LCH ₄ : 80%		PCB for LCH ₄ : 300 ± 50 kPa

The CBH tank was insulated in a vacuum chamber, which indicates the heat conduction and natural convection heat transfer through air can be ignored. Therefore, the main pathways of heat leakage from the hot-end environment to the CBH tank are heat radiation through the multi-layer insulation (MLI) and heat conduction through penetrating components (penetrative heat leakage). This study assumes that the heat leakage from the hot-end to the tank consists of penetrative heat leakage and MLI heat leakage, with other factors being neglected. Based on the experimental system, a schematic diagram of the thermal resistance network between the environment and the tank is shown in Figure 3, and the advanced layer-by-layer model and Fourier's law are used to calculate the heat leakage.

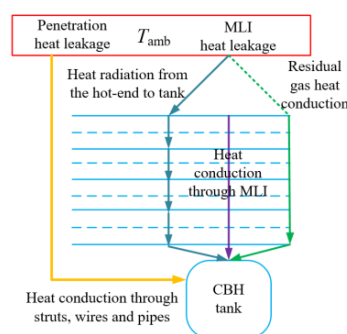


Figure 3. Schematic diagram of the thermal resistance network.

3. Results and discussion

3.1 Heat leakage and storage state

The temperature stratifications in the CBH tank using different schemes are presented in Figure 4. During the storage process, the storage temperature of LOX is close to LCH₄ stored in another compartment, which is termed "synthermal common bulkhead". Due to helium-methane mixture exhibits a higher thermal conductivity than that of methane, the heat transfer at the gas-liquid interface of LCH₄ and the thermal stratification in the LCH₄ compartment are enhanced. Compared with the no-helium scheme, the average temperature, and the temperature gradient in the LCH₄ compartment in the helium-mixing scheme increase by 1.13 K and 8.94%, respectively. The results indicate that helium gas has a minor impact on the LOX compartment, whereas it causes a temperature rise in the LCH₄ compartment and an increment of temperature difference between the two compartments of the tank. Furthermore, regardless of whether helium gas is introduced, distinct thermal stratification occurs in the liquid oxygen region inside the LOX compartment of the tank.

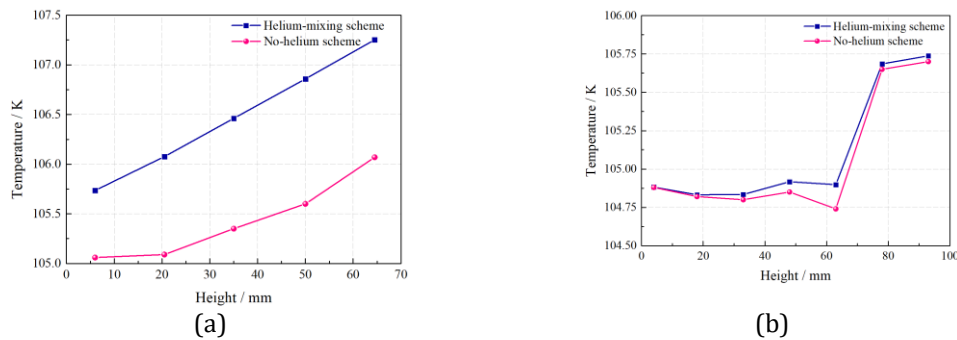


Figure 4. Temperature stratification in the CBH tank. (a) LCH₄ compartment. (b) LOX compartment.

Figure 5 shows the comparison of the heat leakage from the hot end to the CBH tank, and the storage pressures inside the CBH tank using different scheme. The results show the subatmospheric pressure inside the LCH₄ compartment is eliminated owing to an appropriate amount helium gas is introduced into the LCH₄ compartment. In addition, a significant increase of the heat leakage after introducing helium gas into the LCH₄ compartment. Compared with the no-helium scheme, the heat leakage increases by 4.09 W. According to the thermophysical properties of helium-methane mixture, the heat leakage through the residual gas inside the filling and venting pipelines has minor effect on the increase of the total heat leakage, which implies that the introduced helium has more complex influence on the synthermal common bulkhead storage of LOX and LCH₄.

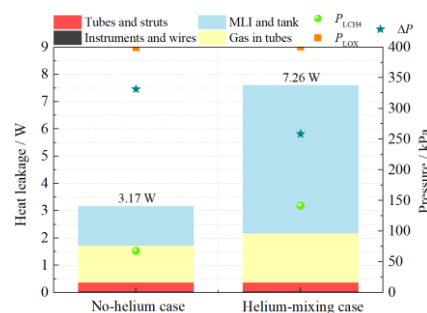


Figure 5. Comparison of the heat leakage and storage pressure.

3.2 Self-pressurization and venting process

Storage duration of cryogenic propellant is vital to evaluate the performance of a cryogenic storage scheme. During the self-pressurization and venting process, the total storage duration is defined as the period in which the filling ratio gradually decreases from its initial value to zero. The self-pressurization and venting behaviors of LOX and LCH₄ in the no-helium scheme and helium-mixing scheme are illustrated in Figure 6. Owing to the thermal conductivity of helium-methane mixture is higher than that of methane, the initial self-pressurization durations in the LOX and LCH₄ compartments in the helium-mixing scheme are shorter and the venting behavior occurs earlier than those in the no-helium scheme. Figures 6 (a) and (b) compare the oxygen self-pressurization and venting behaviors when using different schemes; Figures 6 (c) and (d) compare the methane self-pressurization and venting behaviors. In the no-helium scheme, the temperatures in the LOX compartment and the LCH₄ compartment are higher than those of the helium-mixing scheme. In contrast to the no-helium scheme, the helium-mixing scheme effectively mitigates the temperature fluctuations in the LCH₄ compartment through frequent oxygen venting.

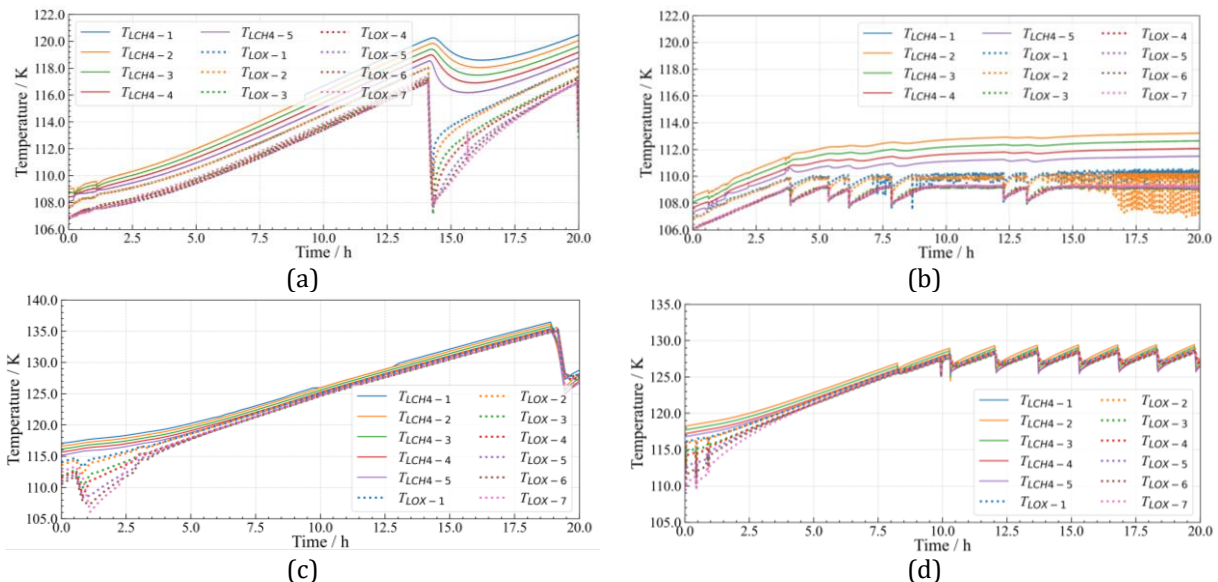


Figure 6. Pressure variation during the self-pressurization and venting process. (a) LOX self-pressurization and venting process of the no-helium scheme. (b) LOX self-pressurization and venting process of the helium-mixing scheme. (c) LCH₄ self-pressurization and venting process of the no-helium scheme. (d) LCH₄ self-pressurization and venting process of the helium-mixing scheme.

Regardless of the scheme, methane mass loss occurs only after the completion of the oxygen self-pressurization and venting, which indicates that LOX evidently plays a protective role against LCH₄ venting during the common bulkhead storage process. Figure 7(a) summarizes the storage durations and venting frequency of oxygen in different schemes. Due to the influence of the methane-helium mixture in the gas region of the LCH₄ compartment, the internal heat leakage through the CBH from the LCH₄ compartment to the LOX compartment enhances, which results in a significant rise in the frequency of venting. Figure 7(b) presents the mass loss during the storage processes in different schemes. In comparison to the no-helium scheme, although helium mixing causes earlier mass loss for LOX, the mass loss rate is lower. Therefore, the storage durations of the helium-mixing scheme are extended. Furthermore, in both schemes, the mass loss of LCH₄ occurs only after the completion of the oxygen self-pressurization and venting process, which indicates that LOX stored in the nonadiabatic CBH tank evidently plays a protective

role against LCH_4 venting. According to Figure 7, in the helium mixing scheme, the self-pressurization venting storage durations for LOX and LCH_4 increase by 38.30% and 9.77%, respectively.

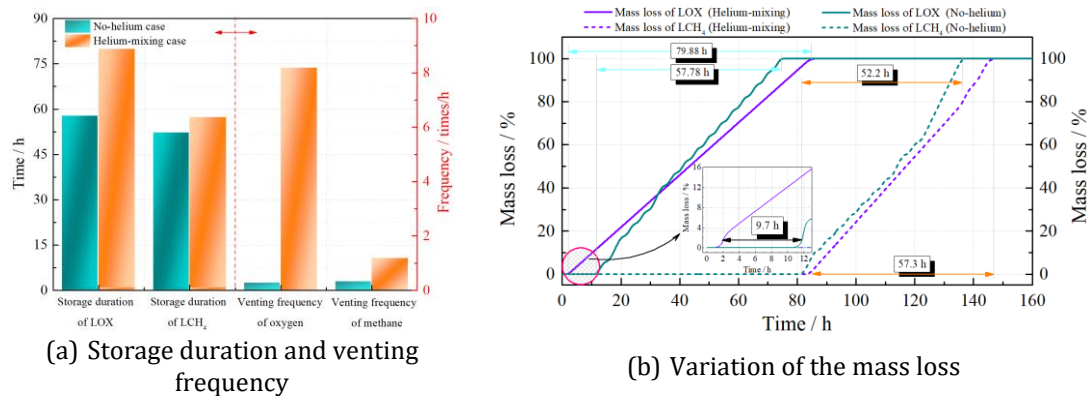


Figure 7. Storage performance during the self-pressurization and venting process.

4. Conclusions

A helium-mixing scheme is proposed to improve the pressure performance in a nonadiabatic common bulkhead tank designed for storing liquid oxygen and liquid methane. The following conclusions can be drawn.

The introduction of an appropriate amount of helium gas into the liquid methane compartment leads to an enhancement in the synthermal common bulkhead storage performance. In the proposed helium-mixing scheme, the subatmospheric pressure in the liquid methane compartment is eliminated with a penalty of a slight thermal performance degrade.

The nonadiabatic common bulkhead contributes to prolonging the total storage duration of liquid methane during the self-pressurization and venting storage process. In comparison to the no-helium scheme, the helium-mixing scheme not only prevents the methane venting behavior but also suppresses the temperature fluctuations in the liquid methane compartment during the liquid oxygen self-pressurization and venting process.

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